

Inference-Time Decoding for Brain Metastasis Segmentation: Ensembling, Hysteresis Thresholding, and Post-processing Validation

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Abstract. Lesion-wise evaluation makes brain metastasis segmentation a joint detection and delineation problem: lesions contribute equally regardless of volume, while unmatched predictions reduce lesion-wise overlap scores and detection F_1 . We investigate how inference-time decoding affects a standard nnU-Net ResEnc-L system. An unweighted average of three trained models followed by hysteresis thresholding increased lesion-wise Dice from 0.560, 0.622 and 0.593 to 0.725, 0.766 and 0.746 for WT, TC and ET, respectively, with these two steps accounting for more than four fifths of the observed improvement. Hysteresis uses a high seed threshold to decide whether a component exists and a lower threshold to recover its extent, without filtering by component size. On 80 training cases, false-positive components had higher peak probabilities than missed reference lesions (median 0.96 versus 0.55), and seven tested suppression strategies based on component size or confidence failed to improve detection without sacrificing true lesions. We also found that post-processing selected on cases used for training could transfer poorly when it altered component retention: a cavity rule estimated at +0.028 Dice locally produced -0.016 on official validation. These results provide a practical inference-time recipe and caution against treating training-set post-processing experiments as evidence of held-out improvement.

Keywords: Brain metastases · Lesion-wise evaluation · Ensembling · Hysteresis thresholding · False positives · nnU-Net · BraTS.

1 Introduction

Brain metastases often present as numerous, small and spatially scattered targets, making their automated delineation substantially different from whole-region tumour segmentation. Earlier glioma benchmarks that shaped the field

scored whole regions [1], whereas BraTS-METS uses lesion-wise evaluation [2,3,4]. Predicted components are matched one-to-one to reference components [11]; each matched lesion contributes equally to the Dice and normalised surface Dice (NSD) averages irrespective of volume, while unmatched predictions receive zero overlap and count as false-positive detections under the applicable small-instance evaluation.

This inverts the usual economics. Under volumetric Dice a five-voxel false positive is negligible; under lesion-wise scoring it costs as much as missing a large lesion. Conversely, a method that suppresses small components to protect Dice can damage the detection metric the task is intended to measure. Over the operating range we tested, improvements in detection and delineation frequently traded off, as quantified in Section 3.3.

We began from a standard nnU-Net ResEnc-L [6,7] baseline and found repeatedly that changes to the network bought less than changes to how its probabilities became a label map. Our contributions are:

1. **An effective inference-time recipe.** Ensembling as a false-positive dilution mechanism, followed by hysteresis decoding, contributes more than four fifths of the total Dice improvement over our baseline (Section 3.2) without retraining the constituent models.
2. **An analysis of the residual error.** The remaining false positives are confident, coherent, isolated and corroborated across independently trained networks. Across seven tested suppression strategies, simple component-size and confidence criteria did not separate them from true lesions in our models (Section 4.1).
3. **An empirical warning for lesion-wise post-processing.** We show how tuning on cases used for training can conceal the false-positive cost of rules that retain additional components, and quantify one reversal between local and official validation (Section 4.4).

What is and is not novel. No individual component is new: nnU-Net and its residual-encoder variants are established [6,7], probability averaging is standard, hysteresis dates to Canny [9], and our third member’s objective follows blob loss [8,10]. The contribution is therefore an empirical analysis of where performance arose under lesion-wise scoring, together with practical lessons from the resulting error analysis and validation failures.

2 Methods

2.1 Data

We use the BraTS-METS 2026 training set [3,4]: 1295 multi-site cases, each with four co-registered, skull-stripped sequences (T1, T1-Gd, T2, T2-FLAIR) supplied in *native* space, so voxel spacing and array dimensions vary across the cohort. Reference labels comprise non-enhancing tumour core (NETC), surrounding non-enhancing FLAIR hyperintensity (SNFH), enhancing tumour (ET) and

Table 1. Ensemble members. All are nnU-Net ResEnc-L, `3d_fullres`, trained on all data.

member	data configuration	training objective	role in the ensemble
TopK	4-region	Dice + top- K BCE	high-precision base
completion	6-region	Dice + top- K BCE	decorrelated second view
CC-loss	6-region	component-weighted fine-tune	high-recall detector

resection cavity (RC). Region prevalence is markedly imbalanced (54% NETC, 89% SNFH, 96% ET and only 13% RC), which is the principal reason the RC channel is our weakest and is handled separately below. Validation uses the official 179-case set scored on the challenge server, which runs the organisers’ evaluation containers through the MedPerf platform [5]; we never see its reference labels.

We predict four *overlapping* binary regions with independent sigmoids rather than mutually exclusive classes,

$$\text{WT} = \{1, 2, 3\}, \quad \text{TC} = \{1, 3\}, \quad \text{ET} = \{3\}, \quad \text{RC} = \{4\},$$

decoded in the order $\text{WT} \rightarrow \text{TC} \rightarrow \text{ET} \rightarrow \text{RC}$ so nested regions overwrite their parents. This is the `Dataset007` configuration in our released code.

2.2 Ensemble members

The submission averages three ResEnc-L models sharing the residual-encoder plans and the `3d_fullres` configuration (Table 1), all trained on all available data without a held-out fold, a decision whose consequences we return to in Section 4.4.

The third member is fine-tuned with a connected-component-aware objective that weights each lesion equally rather than each voxel, following the blob-loss principle [8,10]. It is our strongest detector and our weakest segmenter. Submitted alone it reached small-instance F_1 of 0.429/0.530/0.543, the highest we ever recorded, while its lesion-wise Dice fell to 0.670/0.715/0.691, below the final ensemble; used alone, its additional false positives offset its detection gains. This tension is precisely what the ensemble exploits.

Inference uses sliding-window prediction with `step_size` 0.25 and test-time augmentation by mirroring, retaining per-region probability maps.

2.3 Ensembling as false-positive dilution

The tumour probability map is the unweighted mean over members m ,

$$\bar{p}_r(x) = \frac{1}{3} \sum_{m=1}^3 p_r^{(m)}(x), \quad r \in \{\text{WT}, \text{TC}, \text{ET}\}.$$

The mechanism is asymmetric, and the asymmetry is the point. A detection supported by all three members retains a high mean; one idiosyncratic to a single member is divided by three and falls below the decision threshold. Averaging the high-recall CC-loss member into two cleaner models reduced false positives from **1.66 to 0.85 per case** on the local 80-case harness while retaining its corroborated true detections. Dice and F_1 rose *together* rather than trading in this comparison, making the ensemble more than a variance-reduction convenience in our system.

The resection cavity is treated separately. RC was typically a single large structure in our data, and ensembling gave no measurable benefit: a two-model RC average moved lesion-wise Dice from 0.630 to 0.628. We therefore take RC from the CC-loss member alone, threshold at 0.5, and retain only the largest connected component.

2.4 Hysteresis decoding

Converting $\bar{p}$ to a mask with one global threshold t forces a single number to answer two unrelated questions. A low t promotes every noise speck to a lesion; a high t amputates every real lesion at its margin, because the tumour boundary in these probability maps lies between roughly 0.30 and 0.60. No single value is correct for both.

Hysteresis thresholding, borrowed from classical edge detection [9], splits the two decisions across thresholds $t_{lo} < t_{hi}$. Writing $\mathcal{C}(\cdot)$ for the set of 26-connected components of a binary mask, the decoded region is

$$\hat{M} = \bigcup \left\{ c \in \mathcal{C}(\bar{p}_r > t_{lo}) : \max_{x \in c} \bar{p}_r(x) > t_{hi} \right\},$$

that is, every component of the permissive mask whose *peak* clears the strict threshold is retained *at its full permissive extent*. Figure 1 illustrates the three cases that matter.

The critical property is what the rule does *not* use. It thresholds on peak probability and never on component size or mean probability. This is what separates it from the component filters that fail here (Section 4.1): size and mean confidence both correlate with *being a small true lesion*, so filtering on them preferentially destroys the targets the task rewards, whereas peak probability does not. A thirty-voxel metastasis with a single voxel at 0.9 survives intact; a large diffuse region never reaching 0.7 anywhere is removed in full. Under this decoding rule, one confident voxel is sufficient to retain a component, while the lower threshold determines its spatial extent.

Hysteresis is applied per region to WT, TC and ET, and excluded for RC.

2.5 Final pipeline and reproducibility

Both thresholds were selected on the local harness of Section 4.4 and confirmed on the official validation set. All code, the exact decode, the container definition

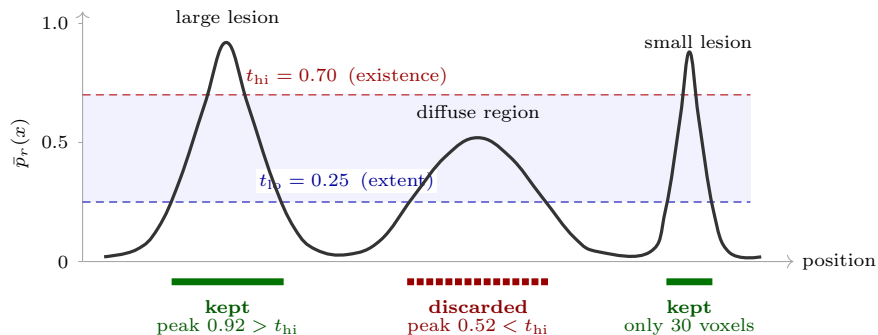


Fig. 1. Hysteresis decoding on a 1-D slice through the fused probability map; the shaded band spans t_{lo} to t_{hi} . Thresholding at t_{hi} alone would keep only the peaks above the band, amputating each lesion’s margin; at t_{lo} alone it would admit the diffuse region. Hysteresis keeps every component of $\bar{p}_r > t_{lo}$ whose peak clears t_{hi} , at full t_{lo} -extent (green), and drops those that never do (red). Size is never consulted, so the 30-voxel lesion survives on one confident voxel while the far larger diffuse region does not.

Table 2. Best validation submission: three-way fusion, hysteresis $t_{lo} = 0.25$, $t_{hi} = 0.70$. Official lesion-wise metrics, 179 cases.

	WT	TC	ET	RC
Lesion-wise Dice	0.725	0.766	0.746	0.630
Lesion-wise NSD	0.741	0.815	0.806	0.521
Small-instance F_1	0.376	0.478	0.476	0.167

and the diagnostic scripts behind every number in this paper are released. Inference is deterministic given the checkpoints; the submitted container bakes in all three and requires approximately 22s per case on the GPU used for submission testing.

3 Results

3.1 Validation performance

Table 2 reports our best submission on the 179-case official validation set.

Performance is markedly uneven across metric families. Overlap and surface metrics are strong, while the three tumour F_1 cells sit far below at 0.376 to 0.478: indicating substantial missed and spurious detections even where matched lesions are reasonably well delineated. Detection is therefore the dominant residual problem for the tumour regions.

On uncertainty and significance. We report point estimates without confidence intervals, which is a genuine limitation rather than an oversight: the challenge

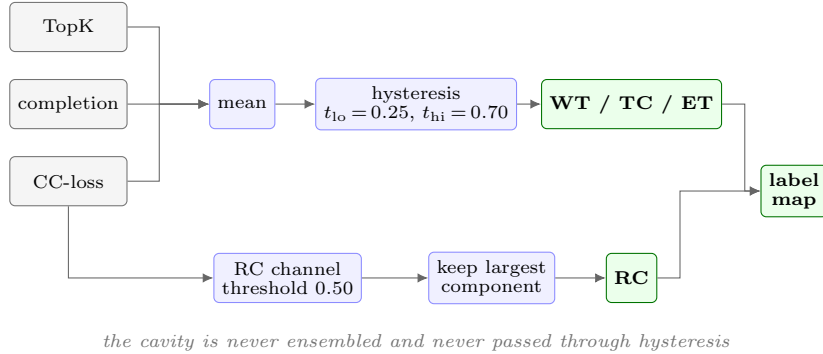


Fig. 2. The submitted pipeline. Tumour regions are the unweighted mean of three members’ probability maps decoded by hysteresis; the cavity comes from the CC-loss member alone. The two compose into one label map by painting WT, TC, ET, then RC, so nested regions overwrite parents.

Table 3. Development path. All rows are our own submissions scored on the official 179-case validation set, and are therefore directly comparable.

change	Dice WT/TC/ET	F_1 WT/TC/ET
1 baseline nnU-Net ResEnc-L	0.560/0.622/0.593	0.379/0.469/0.463
2 + ensembling	0.666/0.708/0.684	0.368/0.439/0.436
3 + second member, step 0.25	0.678/0.718/0.696	0.370/0.477/0.484
4 + RC from the CC-loss member	0.679/0.719/0.696	0.370/0.477/0.484
5 + hysteresis	0.711/0.753/0.729	0.365/0.468/0.475
6 + third member	0.720/0.761/0.737	0.376/0.479/0.477
7 + extent t_{lo} 0.30 $\rightarrow$ 0.25	0.725/0.766/0.746	0.376/0.478/0.476

server returns only aggregate validation scores, and per-case metrics and reference labels are withheld from participants, so dispersion and paired significance tests cannot be computed there at all. Where we control the evaluation, we report per-case counts instead. We therefore avoid treating small aggregate differences as evidence of statistically meaningful superiority.

3.2 Development sequence

This table is a sequential development trace rather than a factorial ablation, so each row-to-row change is conditional on the preceding configuration. Two steps dominate Table 3. Ensembling contributes +0.106/ +0.086/ +0.091 Dice and hysteresis a further +0.032/ +0.034/ +0.033; together they account for +0.138/ +0.120/ +0.124 of the +0.165/ +0.144/ +0.153 total, or over four fifths of it. Both are inference-time operations on frozen probability maps. The tumour F_1 column tells a complementary story: neither step improves it, and the detection gains arrive only with the high-recall third member, which is precisely

Table 4. Fusion weight of the CC-loss member, on the official validation set. Across the tested weights, gains in detection were accompanied by lower Dice.

CC-loss weight	Dice WT/TC/ET	F_1 WT/TC/ET
1/3 (before t_{lo} refinement)	0.720/0.761/0.737	0.376/0.479/0.477
0.42	0.695/0.738/0.714	0.403/0.504/0.510
1/2	0.670/0.715/0.691	0.429/0.530/0.543

the tension Section 3.3 quantifies. Over the same period the components that cost most to develop contributed nothing that survived. A tumour-specialist and a cavity-specialist model were each *worse* than the general network at their own speciality. A distance/boundary loss, Laplacian-of-Gaussian hint channels, MedNeXt-v2, KiU-Net, U-Mamba and larger kernels were all abandoned after failing to beat the baseline. We also determined that “ResEnc-XL” and “ResEnc-L” instantiate an identical network for this dataset, the planners differing only in memory target, so architecture scaling is not a decorrelation lever here.

3.3 The Dice– F_1 trade-off

Interpolating the fusion weight of the high-recall CC-loss member produced a monotone trade-off over the tested weights (Table 4). F_1 can be bought, but only by paying Dice, and the exchange rate is unfavourable: moving from weight 1/3 to 1/2 buys $+0.053/+0.051/+0.066$ F_1 at a cost of $-0.050/-0.046/-0.046$ Dice across the three tumour regions. We retained 1/3 because it gave the best overall balance across the 12 reported region–metric cells among the tested settings; this is not intended to reproduce the organisers’ final ranking procedure.

The same trade separates our two thresholds, which are *not* interchangeable knobs. Lowering the seed threshold t_{hi} from 0.70 to 0.65 gained F_1 $+0.042/+0.039/+0.043$ and lost Dice -0.017 in mean tumour Dice: a large move along the trade-off, and one we rejected. Lowering the extent threshold t_{lo} from 0.30 to 0.25 gained Dice $+0.005/+0.005/+0.009$ for approximately -0.001 mean F_1 : a favourable change over the tested setting, and the configuration we submitted. Pushing further to $t_{lo} = 0.20$ returned to trading, adding $+0.003/+0.002/+0.002$ Dice for -0.003 WT F_1 , which brackets the useful range of the extent threshold at $t_{lo} = 0.25$.

4 Discussion

4.1 Confidence does not separate false positives from missed lesions

On 80 training cases evaluated with the official matching code, the median peak probability of a false-positive component was **0.96**, whereas the median maximum probability within a missed reference lesion was **0.55**. We detected 45% of small lesions; in 43% of the misses, the probability did not exceed 0.10 anywhere

Table 5. Seven suppression strategies evaluated. Across post-hoc, loss-level and seed-threshold interventions, none separated false-positive components from true lesions without a substantial detection cost in our experiments.

mechanism	outcome
<i>(a) Post-hoc component filters: rank on size or mean confidence</i>	
small-component filters	substantially reduced F_1 by removing small true lesions
rejection by mean confidence	increased Dice but substantially reduced recall
<i>(b) Loss-level suppression: rank on a learned per-component score</i>	
per-component FP penalty	FP -53% but TP -24% ; collapsed into global under-confidence
the same model as a veto	at $t = 0.5$ it rejected <i>everything</i> (TP $1.82 \rightarrow 0.51$)
blending the veto with the base	FPs and TPs removed in lockstep at every mixing ratio
<i>(c) Seed-threshold variants: rank on a transformed confidence</i>	
size-adaptive seed threshold	reproduces the plain global threshold exactly
scale-space (smoothed) seed	at every matched FP rate the plain threshold detects more

within the reference component. In addition, 86% of false-positive components were beyond the pre-specified proximity threshold from any reference lesion.

The false-positive components that survive decoding therefore have substantially higher peaks than the reference lesions still missed by the model. They are also spatially coherent (remaining solid after Gaussian smoothing), isolated and often corroborated across the three independently trained networks.

We evaluated seven suppression strategies (Table 5) spanning post-hoc component filtering, loss-level interventions and seed-threshold variants. None provided a useful separation between false positives and true lesions in our experiments. This does not rule out all confidence- or size-aware methods, but it indicates that the simple component-level criteria we tested are poor discriminants for these models. Two failure modes were especially clear: the per-component loss penalty produced global under-confidence, while the learned veto removed true and false components together.

4.2 False positives are consistent with anatomical mimics

Two explanations remained. Either these are anatomical *mimics* (enhancing vessels, choroid plexus, dural structures) that a radiologist excludes using context the network lacks, or they are *real metastases the annotators missed*, in which case part of the ceiling is not ours to move.

Non-expert visual review of the 48 most confident false positives was consistent with the mimic hypothesis. They are overwhelmingly small bright enhancing

foci in three anatomically suspicious locations: at the cortical margin or dura, on or beside bright linear tubular structures consistent with vessels, and at the midline in falx and superior sagittal sinus territory. Few appeared as isolated round parenchymal foci away from conspicuous anatomical structures, although this distinction cannot be established reliably without expert review.

That review was carried out by a non-radiologist on single axial slices, so we treated it as a hypothesis to test rather than a clinical assessment. Anatomical position is a genuinely different separation axis from confidence, so we measured it. Distance from the brain surface to the false-positive centroid has a median of 8.8 mm against 11.2 mm for reference lesion centroids, with almost fully overlapping interquartile ranges and an area under the ROC curve of **0.575**. False positives are only marginally more peripheral, and grey-white junction metastases, a common true presentation, sit at the same depth, so surface proximity alone is unlikely to remove many suspected mimics without also removing real lesions. The measurement is worth reporting precisely because the hypothesis is attractive: anatomical context is the obvious thing a radiologist uses and a network lacks, and this result suggests that any useful anatomical discriminator would need to be substantially more specific than raw surface proximity.

4.3 What target-level intervention buys, and what it costs

Since decoding could not separate these errors, we attacked the 43% of misses to which the network is wholly blind from the training side.

Sampling frequency is not the bottleneck. nnU-Net’s sampler draws a random key from `class_locations` and then a random voxel within it, so a thirty-voxel metastasis sharing a key with every large-lesion and oedema voxel is patch-centred approximately 0.3% of the time. We added a dedicated key containing only small-component voxels, with every lesion weighted equally regardless of size. Verified over 4000 draws, the probability of centring a patch on a small lesion rose from 1.28% to 25.75%, a twenty-fold increase. Small-lesion detection did not improve materially. This suggests that patch-sampling frequency was not the primary cause of the observed misses in this experiment.

Enlarging the target does find more lesions, at a price the metric sets. We instead dilated isolated small targets, so a thirty-voxel lesion presents roughly four times the volume and a larger object for the receptive field, with a matched selective erosion at inference. Submitted alone this lifted tumour F_1 by +0.012/+0.014/+0.013, evidence that a meaningful share of the extra detections are real. Grafting only the additional, non-overlapping detections onto our best submission raised F_1 by as much as +0.034/+0.027/+0.031 but lowered lesion-wise Dice by −0.010/−0.012/−0.012. The overall balance across the reported metrics was negative for our selected operating point. The reason is specific to lesion-wise scoring and worth stating plainly: *many newly detected small lesions entered the per-lesion Dice average at a below-average value*, because small lesions are segmented less accurately than large ones. Under volumetric Dice these small components would receive much less weight; under lesion-wise Dice they contribute

equally. A more aggressive variant added 156 further components averaging six voxels each and bought only $+0.003/ +0.004/ +0.005$ F_1 , indicating that most of these very small additional components were false positives.

4.4 Lessons for lesion-wise post-processing validation

We built a local harness around the official evaluation code using 80 reference tumour cases and 100 cavity cases. Because every model had been trained on these cases, this harness was not an unbiased estimate of generalisation. We used it only as a ranking heuristic and first required it to reproduce the relative ordering of our own previously scored submissions. It then allowed us to compare 28 candidate decodes without consuming additional submissions.

The rankings transferred reasonably for the threshold and fusion settings we tested, but failed for a rule that explicitly retained additional cavity components. On memorised cases, spurious components were uncommon, so the rule appeared to add true positives with little false-positive cost. On unseen validation data, the same rule also admitted spurious components. A cavity keep-rule estimated locally at $+0.028$ lesion-wise Dice instead produced -0.016 on official validation, reversing the sign of the predicted effect.

This observation supports a narrower practical recommendation than a universal protocol. Results from training-set post-processing should be treated as provisional, especially when a rule changes component retention. A genuine held-out fold is preferable; when none exists, at least one official submission should be reserved to validate such changes. Threshold and fusion rankings happened to transfer better in our experiments, but training-set agreement does not guarantee held-out improvement for those settings either.

4.5 Conclusion

Our BraTS 2026 Task 1 submission reaches WT/TC/ET lesion-wise Dice of $0.725/0.766/0.746$ using probability averaging and hysteresis decoding without changing the constituent network architectures. The largest remaining weakness is tumour detection. In our models, simple confidence- and size-based suppression did not separate false-positive components from true lesions, non-expert review suggested that many errors may be anatomical mimics, and attempts to recover additional small lesions traded improved F_1 against lower lesion-wise Dice. The main practical lesson is therefore methodological: inference-time decoding can materially change challenge performance, while post-processing selected on cases used for training must not be treated as evidence of held-out improvement. All code is released at <https://github.com/aymuos15/brats2026-mets>.

Acknowledgements

Data used in this publication were obtained as part of the Challenge project through Synapse ID syn74274097.

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